

Privacy Router: A Sovereignty-First Butler for Korean Researchers

GenAI & Blockchain — Term Project Proposal · DH. Kim & M. Saadati · Type (iv) Privacy-Critical Agent · 2026-05-25

1. Problem and thesis

Slide A.2 of L1: “A Private AI is a routing layer that knows what must stay home.” OPF is US-centric (SSN, ZIP), omitting Korean RRN, +82 phones, and institutional IDs. Agent frameworks also draft emails/autofill forms autonomously, often embedding invisible secrets into external LLM calls. For instance, an agent might silently ship your **RRN** (resident ID) or **unpublished data** to Claude/external LLMs. We build **Privacy Router**: an on-device **Extractor** → **Judge** → **Router** pipeline that intercepts every agent-generated prompt before egress. A Small Language Model (SLM; e.g., Qwen3.6 9B) detects secrets the user overlooks; a policy-driven Judge then determines if the prompt is safe for external APIs or must remain local.

2. System architecture

Three layers, one custom MCP server in the middle.

Extractor. *Layer 1* (pattern) detects RRN, +82 phones, and IDs via a credential-scanner MCP server. *Layer 2* (context) uses the SLM to detect research/business secrets from session memory. Both layers emit structured audit records.

Judge. Consumes records, applies a configurable YAML/JSON policy schema, and decides domain (personal / business / research), sensitivity (sensitive / mixed / non-sensitive), and action (auto_redact, prompt_user, block). It emits a structured judgment record, with the final label being model-driven.

Router. *Maskable* (mixed class): replaces sensitive spans with placeholders ([PHONE_NUMBER#1], [RRN#1]) before egress to Claude Haiku 4.5 via OpenRouter, then rehydrates the response. *Local-only* (sensitive or user-declined masking): processed by the local Ollama/llama.cpp worker (qwen3:14b). Orchestrator escalation (Opus 4.7) is reserved for planner-flagged hard reasoning.

Application: The router intercepts all agent-generated prompts at the MCP boundary, ensuring safety by default, and is invoked from Telegram, Slack, and CLI.

3. MCP plan

Three free servers + two custom servers.

filesystem (sandboxed ~/workspace) — read paper PDFs, draft notes locally; sensitive content never leaves the sandbox

fetch (uvx mcp-server-fetch) — pull arXiv abstracts and public regulation pages; only invoked on non-sensitive class.

memory — persist routing decisions and curated notes across sessions; small, auditable local SQLite KV store.

credential-scanner (custom MCP, stdio) — lightweight regex + tiny SLM span detector. Emits credential spans (kr_rrn, kr_phone, institutional_id, etc.). Called by the Extractor as a reference tool.

privacy-router (custom MCP, FastMCP, stdio) — three tools: (1) `classify(text: str) → {class, domain, rationale, tool_calls}` — runs Extractor/Judge to return model-generated labels, reasoning, and regex spans. (2) `redact(text: str) → str` — strips matched spans before global LLM calls (mixed class). (3) `route(text: str) → {model, endpoint, rationale}` — returns the chosen worker/orchestrator + audit-log entry.

Threat model. filesystem: reads only ~/workspace (no /, no ~); injected content is untrusted. fetch: confirms on first new domain; results are never auto-acted upon. memory: local SQLite, no remote write. privacy-router: classification is model-driven; the Judge retains final authority and prevents misclassification via regex tampering.

First USAGE_LOG.md entry. 2026-05-24 22:10 CLI privacy-router classify("...010-XXXX-XXXX...") → sensitive, personal (KR-phone match); route → local qwen3:14b; OK, ~340 ms.

4. Three OpenClaw advantages addressed

Privacy by default — the router enforces the Sensitive/Mixed/Non-sensitive split before egress; all decisions are locally audited. **Cost transparency** — two-tier stack with token meter; targets ≤ \$10/month/user, with ~95% of calls on Haiku or local. **Customization** — the Korean-credential regex layer and personal/business taxonomy ship as a versioned skill; SOUL.md defines the router’s butler identity.

5. Future work (out of scope this term)

Replace the regex classifier with an OPF-style fine-tuned model (trained on Korean credentials); integrate with an information system for local retention recommendations based on routing logs.